



Reducing cost per analysis for Kjeldahl

**A Resource-Efficient Approach to Kjeldahl Nitrogen Determination
Using BUCHI's Reaction Detection Sensor® Technology**

INTRODUCTION

The Kjeldahl method is an official worldwide standard that has been approved by various scientific associations including AOAC International, AACC International, AOCS, DIN, EPA, ISO, JAS, and USDA. It is mainly applied in the food and beverage industry but also used in environmental analysis, feed, pharma and the agricultural industry for the determination of nitrogen, protein, nitrates and ammonium. Due to the popularity of this method and its wide range of applications, there is a great demand to make the analysis more cost-effective.

In this Application Note, several Kjeldahl measurements are presented to show the potential of the Reaction Detection Sensor (RDS) technology in saving sodium hydroxide and therefore, lowering the costs of the analysis and making it more environmentally friendly.

The Kjeldahl method is enhanced by the Reaction Detection Sensor (RDS) technology, which significantly reduces sodium hydroxide usage and promoting cost efficiency and environmental sustainability. Traditional methods often use an excess of sodium hydroxide for alkalization, while the RDS technology optimizes this by accurately detecting the alkalization endpoint for a more environmentally friendly analysis.

Keywords or phrases:

Reaction Detection Sensor (RDS)–assisted Kjeldahl analysis, reduction of sodium hydroxide consumption, cost-per-analysis optimization in Kjeldahl distillation, environmentally sustainable nitrogen determination, automated alkalization endpoint detection, high-throughput Kjeldahl protein analysis, resource-efficient Kjeldahl distillation and titration

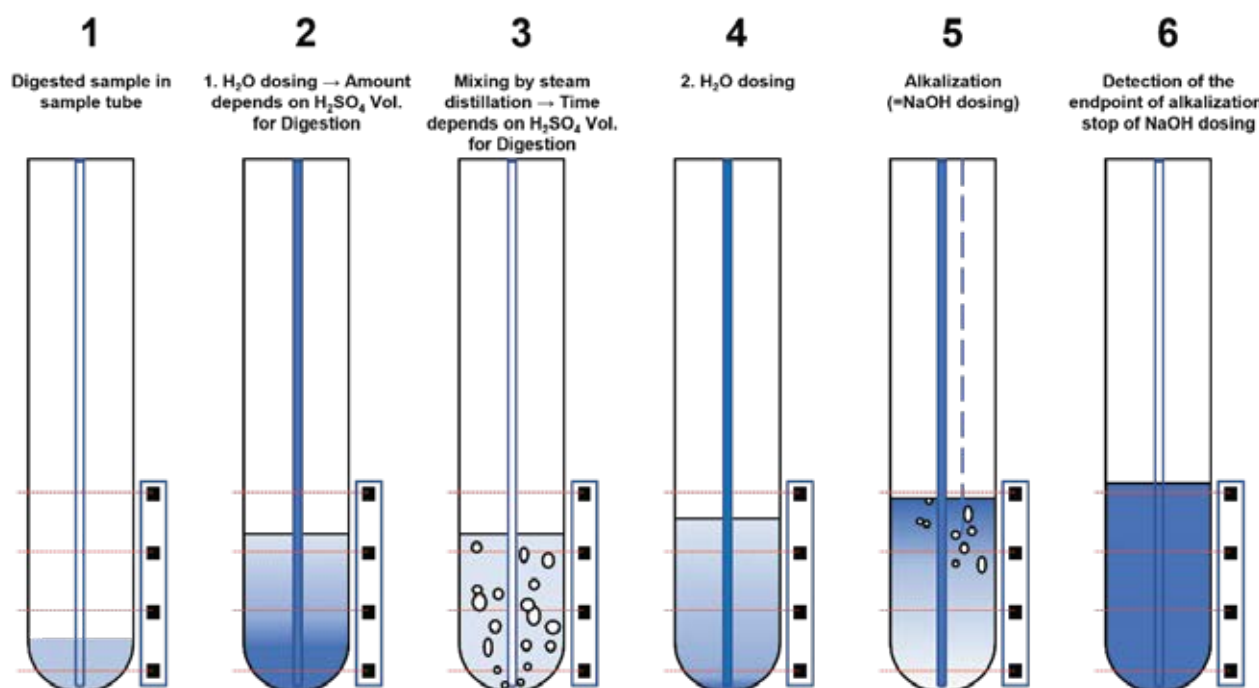


Figure 1: Functional principle of the Reaction Detection Sensor technology.

The Kjeldahl method starts with a catalytically supported mineralization of organic material in a boiling mixture of sulfuric acid and sulfate salts. During this step, the organic nitrogen is converted into ammonium sulfate. In the next step, the solution is alkalinized to liberate the ammonia which is further steam-distilled and determined by titration.^[3] Mineralization of the sample consumes sulfuric acid to form carbon dioxide, water and sulfur dioxide. The more organic matter in the sample, the more sulfuric acid is consumed, and the less acid remains in the sample tube.

In a classical Kjeldahl distillation, the dosed volume of sodium hydroxide solution is always the same and is based on the maximum possible amount of sulfuric acid after mineralization. It is generally dosed in excess to ensure completion of alkalization, which is crucial to release the ammonia and thereby quantify the protein content accurately. If a Kjeldahl distillation is performed with the RDS, the end of the alkalization is detected, and dosage of sodium hydroxide is stopped immediately. Therefore, less sodium hydroxide is dosed to samples that contain less sulfuric acid.

Figure 1 shows the functional principle of the RDS technology.

EQUIPMENT

- ▶ KjelDigester K-449 (1154492000).
- ▶ Scrubber K-415 TripleScrub ECO (114152331).
- ▶ MultiDist coupled with Metrohm EcoTitrator and Recirculating Chiller F-314 (11K36532211).
- ▶ Reaction Detection Sensor (11072666).
- ▶ Sample Tubes 300 mL (037377).
- ▶ BUCHI Mixer B-400 (034220).
- ▶ Analytical balance (accuracy ± 0.1 mg).

CHEMICALS AND MATERIALS

Chemicals:

- ▶ Sulfuric acid conc. 96%, VWR
- ▶ BUCHI Kjeldahl Tablets Titanium, Missouri, ECO & Antifoam.

- ▶ Sodium hydroxide 32%, VWR.
- ▶ BUCHI ready to use 4% boric acid pH 4.65 with Sher indicator.
- ▶ Sulfuric acid 0.1 mol/L, VWR.
- ▶ Neutralization solution for the Scrubber: 600 g sodium carbonate, calcined, technical, Synopharm and a spatula tip of bromthymol blue, Fluka (18460) diluted to 3 L with distilled water.
- ▶ Glycine, assay 99.7%, Sigma aldrich.
- ▶ BUCHI Weighing boats.

For safe handling please refer to all corresponding MSDS.

Sample name	Sample details	Declared / labeled protein content (g/100g)
Milk powder	Reference material from LVU (Nr. 18-4a)	20.38 $\pm$ 0.37
Emmentaler mild	From Supermarket	29
Boiled sausage	Reference material from LVU (Nr. 20-01ab)	16.19 $\pm$ 0.20
Milk chocolate	From Supermarket	7.7

Table 1: Sample details.

PROCEDURE

The determination of nitrogen and protein in various food samples includes the following steps:

- ▶ Sample homogenization
- ▶ Digestion of the sample
- ▶ Distillation and titration

SAMPLE HOMOGENIZATION

The milk powder sample was used directly for analysis. The boiled sausage sample was ground to a fine paste using a mixer. The Emmentaler cheese sample as well as the milk chocolate sample were cut into small pieces ($\varnothing$ 2–3 mm).

DIGESTION OF THE SAMPLE

- ▶ After sample homogenization, digestion was carried out as follows:
- ▶ Samples were weighed in BUCHI weighing boats with nitrogen free paper. Each weighed portion of sample was placed in a 300 mL Sample Tube.
- ▶ The Kjeldahl/Antifoam Tablets and the corresponding volume of sulfuric acid (conc. 96%) were added.

- ▶ Additional blanks (chemicals without the sample) as well as reference samples (Glycine) were prepared.
- ▶ The racks containing the blanks and the samples were inserted into the Digestion Unit (KjelDigester K-449) and the suction module was closed.
- ▶ The digestion was started.

Table 2, 3 and 4 show an overview of the samples that were analyzed in the course of this study. The corresponding digestion methods are shown in Table 5, 6 and 7.

No.	Type	Kjeldahl Tablets type	Kjeldahl Tablets [pcs.]	Antifoam Tablets [pcs.]	Conc. H ₂ SO ₄ [mL]	Sample weight [g]
1–5	Blank	Titanium	1	1	8	–
6–10	Reference (Glycine, 99.7% purity)					0.09–0.10 g
11–15	No. 1: Milk powder					0.4–0.5 g
15–20	No. 2: Emmentaler					0.8–1.0 g
21–25	No. 3: Boiled sausage					1.6–1.8 g
26–30	No. 4: Milk chocolate					0.6–0.7 g
31–35	Blank		2	1	15	–
36–40	Reference (Glycine, 99.7% purity)					0.09–0.10 g
41–45	No. 1: Milk powder					0.4–0.5 g
46–50	No. 2: Emmentaler					0.9–1.0 g
51–55	No. 3: Boiled sausage					1.8–2.0 g
56–60	No. 4: Milk chocolate					1.0–1.1 g

Table 2: Overview of the samples that were digested using Kjeldahl Tablets Titanium.

No.	Type	Kjeldahl Tablets Type	Kjeldahl Tablets [pcs.]	Antifoam Tablets [pcs.]	Conc. H ₂ SO ₄ [mL]	Sample weight [g]
61–65	Blank	Missouri	1	1	10	–
66–70	Reference (Glycine, 99.7% purity)					0.09–0.10 g
71–75	No. 1: Milk powder					0.4–0.5 g
76–80	No. 2: Emmentaler					0.8–1.0 g
81–85	No. 3: Boiled sausage					1.6–1.8 g
86–90	No. 4: Milk chocolate					0.6–0.7 g
91–95	Blank		2	1	20	–
96–100	Reference (Glycine, 99.7% purity)					0.09–0.10 g
101–105	No. 1: Milk powder					1.0–1.2 g
106–110	No. 2: Emmentaler					0.9–1.0 g
111–115	No. 3: Boiled sausage					1.8–2.0 g
116–120	No. 4: Milk chocolate					1.2–1.4 g

Table 3: Overview of the samples that were digested using Kjeldahl Tablets Missouri.

No.	Type	Kjeldahl Tablets Type	Kjeldahl Tablets [pcs.]	Antifoam Tablets [pcs.]	Conc. H ₂ SO ₄ [mL]	Sample weight [g]
121–125	Blank	ECO	1	1	8	-
126–130	Reference (Glycine, 99.7% purity)					0.09–0.10 g
131–135	No. 1: Milk powder					0.4–0.5 g
136–140	No. 2: Emmentaler					0.8–1.0 g
141–145	No. 3: Boiled sausage					1.6–1.8 g
146–150	No. 4: Milk chocolate					0.6–0.7 g
151–155	Blank		2	1	16	-
156–160	Reference (Glycine, 99.7% purity)					0.09–0.10 g
161–165	No. 1: Milk powder					1.0–1.2 g
166–170	No. 2: Emmentaler					0.9–1.0 g
171–175	No. 3: Boiled sausage					1.8–2.0 g
176–180	No. 4: Milk chocolate					1.2–1.4 g

Table 4: Overview of the samples that were digested using Kjeldahl Tablets ECO.

Step	Temperature [°C]	Time [min]
1	250	0
2	280	20
3	380	30
4	420	80
Cooling	-	35

Table 5: Digestion parameters for samples no. 1–60 (Kjeldahl Tablets Titanium, 8 & 15 mL of conc. H₂SO₄).

Step	Temperature [°C]	Time [min]
1	250	0
2	280	20
3	380	30
4	420	90
Cooling	-	35

Table 6: Digestion parameters for samples no. 61–120 (Kjeldahl Tablet Missouri, 10 & 20 mL of conc. H₂SO₄).

Step	Temperature [°C]	Time [min]
1	250	0
2	280	20
3	380	30
4	420	100
Cooling	-	35

Table 7: Digestion parameters for samples no. 121–180 (Kjeldahl Tablet ECO, 8 & 16 mL of conc. H₂SO₄).

NOTE: As a first indication of successful digestion, the digested sample should be clear and blue green, with acceptable recoveries of a reference substance.

DISTILLATION AND TITRATION

After digestion, the samples were analyzed as follows:

- ▶ Alkalization of the sample to free up the nitrogen (ammonium).
- ▶ Steam distillation of the ammonia into boric acid receiver.
- ▶ Determination of the nitrogen content by boric acid titration.

Boric acid titration was carried out using sulfuric acid as titrant. An excess of boric acid guaranteed that all ammonia was bound in a complex. Table 8 shows the distillation and titration parameters. Before sample distillation, a priming was performed with the method used for sample analysis.

Method parameters MultiDist	
Reaction Detection	On
H ₂ SO ₄ Vol. for digestion	* mL
Steam steps	No
Steam power	1
Level detection	Off
Distillation time	180 s
Stirrer speed distillation	5
Titration type	Boric Acid Titration
H ₃ BO ₃ Volume	60 mL (4%)
Stirrer speed titration	8
Titration start time	160 s
Sample tube aspiration	30 s
Receiver aspiration	25 s

Instrument Settings	
MaxAccuracy mode	On
Chiller / Tap water	Chiller F-314
Chiller temperature	15 °C
AutoDist mode	On
Titration model	Metrohm Eco Titrator

Automated Titration with Eco Titrator	
Method Eco Titrator	BUCHI blank; BUCHI sample
Titration	H SO 0.1 mol/L
Sensor type	Potentiometric (pH)
Endpoint pH	4.65

Table 8: Parameters for distillation and titration with MultiDist (samples no. 1–180).

*Sample No. 1–30: 8 mL

Sample No. 31–60: 15 mL

Sample No. 61–90: 10 mL

Sample No. 91–120: 20 mL

Sample No. 121–150: 8 mL

Sample No. 151–180: 16 mL

CALCULATION

The results were calculated as a percentage of nitrogen. In order to calculate the protein content of the sample, the nitrogen content is multiplied with a sample-specific protein factor. The following equations (1), (2), and (3) were used to calculate the results. For the reference substance, the purity of the Glycine is considered in equation (4).

$$wN = \frac{(V_{\text{Sample}} - V_{\text{Blank}}) \cdot Z \cdot c \cdot f \cdot MN}{m_{\text{Sample}} \cdot 1000} \quad (1)$$

$$\%N = wN \cdot 100\% \quad (2)$$

$$\%P = wN \cdot PF \cdot 100\% \quad (3)$$

$$\%N_{\text{Gly}} = \frac{\%N \cdot 100}{P} \quad (4)$$

wN: Weight fraction of nitrogen.

 V_{Sample} : Amount of titrant for the sample [mL]. V_{Blank} : Mean amount of titrant for the blank [mL].z: Molar valence factor (1 for HCl, 2 for H₂SO₄).

c: Titrant concentration [mol/L].

f: Titre value (for commercial solutions normally 1.000; refer to the product certificate).

MN: Molecular weight of nitrogen (14.007 g/mol).

 m_{Sample} : Sample weight [g].

1000: Conversion factor [mL to L].

%N: Percentage weight of nitrogen.

 $\%N_{\text{Gly}}$: Percentage weight of nitrogen corrected for the purity of reference substance glycine [%].

%P: Percentage weight of protein.

P: Purity of the reference substance Glycine [%] as declared by the manufacturer.

PF: Sample-specific protein factor (sample type no. 1 & 2 / milk powder & Emmentaler: 6.38; sample type no. 3 & 4 / boiled sausage & milk chocolate: 6.25).

RESULTS

Measurements performed using Kjeldahl Tablets Titanium & Antifoam Tablets

Table 9 and 10 show the results of the samples that were digested using Kjeldahl Tablets Titanium & Antifoam Tablets. The relative standard deviations (RSD) of the blanks (required: < 5%) and the samples (required: < 1%) are fine. Besides that, the recovery rates of the reference samples are

within the required range (98.0–102.0%) and the determined protein contents are very close to the expected values.

Compared to the standard method (KjelOptimizer App), up to Ø 43.0% of NaOH per sample was saved for the first measurement series due to the RDS technology (Table 9). In total, 35.3% (=402 mL) of NaOH was saved. For the second measurement series, up to Ø 34.3% of NaOH per sample and 19.0% (= 359 mL) for the whole measurement series was saved (Table 10, p. 9).

Type		Titrated volume [mL]		N [%]		Ø Protein [%]		NaOH Volume [mL]		NaOH Vol. saved vs standard method [%]
Blank	-	0.313	0.318	-				28	29.2 ± 2.4	23.2
		0.320						30		
		0.321						27		
		0.320						28		
		0.315						33		
Ref (Gly)	0.096	6.772		18.81	-	-	101.120	29	29.4 ± 0.9	22.6
	0.094	6.606		18.70			100.500	29		
	0.091	6.397		18.69			100.460	31		
	0.096	6.708		18.72			100.640	29		
	0.099	6.883		18.69			100.460	29		
Sample No. 1	0.460	5.634		3.24	20.69	20.70 ± 0.04	-	22	22.0 ± 0.7	42.1
	0.458	5.624		3.25	20.73			21		
	0.479	5.860		3.25	20.72			23		
	0.487	5.951		3.25	20.73			22		
	0.487	5.934		3.24	20.64			22		
Sample No. 2	0.835	14.938	4.91	31.36	31.32 ± 0.17	-	25	22.6 ± 2.8	40.5	
	0.819	14.709	4.93	31.46			18			
	0.894	15.793	4.86	31.00			22			
	0.891	15.898	4.91	31.31			24			
	0.900	16.137	4.94	31.49			24			
Sample No. 3	1.693	16.179	2.63	16.43	16.41 ± 0.07	-	22	22.8 ± 0.8	40.0	
	1.634	15.579	2.62	16.39			22			
	1.641	15.632	2.62	16.37			23			
	1.672	15.858	2.61	16.31			24			
	1.705	16.370	2.64	16.52			23			
Sample No. 4	0.607	3.063	1.27	7.93	7.91 ± 0.01	-	20	21.6 ± 1.5	43.2	
	0.608	3.057	1.26	7.90			22			
	0.613	3.076	1.26	7.90			21			
	0.609	3.059	1.26	7.90			21			
	0.640	3.201	1.26	7.90			24			
Total NaOH Volume [mL]								738		
Expected NaOH Volume without RDS (38 mL/sample) [mL]								1140		
NaOH Volume saved [mL]								402		
NaOH saved [%]								35		

Table 9: Results of the measurements performed using 8 mL of conc. H₂SO₄, Kjeldahl Tablets Titanium & Antifoam Tablets (No. 1–30; RSD: blank = 1.00%; ref = 0.25%; s 1 = 0.17%; s 2 = 0.55%; s 3 = 0.43%; s 4 = 0.17%).

Type		Titrated volume [mL]		N [%]		Ø Protein [%]		NaOH Volume [mL]		NaOH Vol. saved vs standard method [%]
Blank	-	0.2640	0.261	-				60	59.6 ± 0.5	5.4
		0.2600						60		
		0.2600						59		
		0.2610						59		
		0.2620						60		
Ref (Gly)	0.0945	6.5690	18.81	-	-	100.71	58	58.2 ± 0.8	7.6	
	0.0930	6.4560				100.50	57			
	0.0933	6.4900	18.69			100.72	59			
	0.0996	6.8620	18.72			99.99	59			
	0.0959	6.6420	18.69			100.39	58			
Sample No. 1	0.4262	5.1570	3.24	20.69	20.58 ± 0.02	-	51	53.4 ± 4.4	15.2	
	0.4761	5.7370	3.25	20.73			52			
	0.4578	5.5230	3.25	20.72			53			
	0.4164	5.0510	3.25	20.73			61			
	0.4230	5.1160	3.24	20.64			50			
Sample No. 2	0.9179	16.3960	4.91	31.36	31.11 ± 0.26	-	47	46.4 ± 1.3	26.3	
	0.9740	16.9710	4.93	31.46			45			
	0.9595	16.9120	4.86	31.00			45			
	0.9436	16.5770	4.91	31.31			48			
	0.9478	16.8280	4.94	31.49			47			
Sample No. 3	1.9265	17.9830	2.63	16.43	16.15 ± 0.03	-	44	47.2 ± 2.3	25.1	
	1.8788	17.5930	2.62	16.39			48			
	1.8711	17.4730	2.62	16.37			50			
	1.9203	17.9590	2.61	16.31			48			
	1.8676	17.4080	2.64	16.52			46			
Sample No.4	1.0941	5.1700	1.27	7.93	7.86 ± 0.01	-	41	41.4 ± 1.7	34.3	
	1.0887	5.1350	1.26	7.90			40			
	1.0291	4.8720	1.26	7.90			44			
	1.0772	5.0840	1.26	7.90			42			
	1.0954	5.1640	1.26	7.90			40			
Total NaOH Volume [mL]								1531		
Expected NaOH Volume without RDS (38 mL/sample) [mL]								1890		
NaOH Volume saved [mL]								359		
NaOH saved [%]								19.0		

Table 10: Results of the measurements performed using 15 mL of conc. H₂SO₄, Kjeldahl Tablets Titanium & Antifoam Tablets (No. 31–60; RSD: blank = 0.57%; ref = 0.27%; s 1 = 0.08%; s 2 = 0.84%; s 3 = 0.17%; s 4 = 0.09%).

Type		Titrated volume [mL]		N [%]		Ø Protein [%]		NaOH Vol. [mL]		Ø NaOH Vol. saved vs standard [%]
Blank	-	0.296	0.2952	-	-	-	-	40	39.2 ± 2.9	12.9
		0.298						44		
		0.294						38		
		0.295						37		
		0.293						37		
Ref (Gly)	0.0913	6.373		18.69	-	-	100.44	40	40.2 ± 0.4	10.7
	0.0924	6.378		18.48			99.32	40		
	0.0973	6.725		18.55			99.7	40		
	0.0962	6.679		18.63			100.12	41		
	0.0919	6.389		18.61			100.05	40		
Sample No. 1	0.4599	5.604	3.24	20.67	20.69 ± 0.03	-	32	31.8 ± 0.8	29.3	
	0.464	5.649	3.24	20.66			31			
	0.4605	5.615	3.24	20.69			33			
	0.4847	5.897	3.24	20.7			31			
	0.4882	5.948	3.25	20.74			32			
Sample No. 2	0.8748	15.152	4.77	30.41	30.63 ± 0.12	-	31	30.2 ± 2.3	32.9	
	0.9465	16.527	4.81	30.71			29			
	0.8722	15.258	4.82	30.72			31			
	0.9158	16.01	4.82	30.73			33			
	0.987	17.155	4.79	30.59			27			
Sample No. 3	1.729	16.489	2.63	16.43	16.33 ± 0.11	-	35	29.0 ± 6.0	35.6	
	1.6854	15.907	2.6	16.25			24			
	1.6772	15.871	2.61	16.29			29			
	1.6708	15.982	2.64	16.47			22			
	1.7831	16.754	2.59	16.19			35			
Sample No. 4	0.6451	3.192	1.26	7.88	7.88 ± 0.01	-	29	28.8 ± 5.1	36	
	0.6619	3.275	1.26	7.9			25			
	0.6306	3.128	1.26	7.88			31			
	0.6707	3.307	1.26	7.88			36			
	0.6243	3.099	1.26	7.88			23			
Total NaOH Volume [mL]								996		
Expected NaOH Volume without RDS (38 mL/sample) [mL]								1350		
NaOH Volume saved [mL]								354		
NaOH saved [%]								26		

Table 11: Results of the measurements performed using 10 mL of conc. H₂SO₄, Kjeldahl Tablets Missouri & Antifoam Tablets (No. 61–90; RSD: blank = 0.58%; ref = 0.38%; s 1 = 0.12%; s 2 = 0.40 %; s 3 = 0.65%; s 4 = 0.10%).

Measurements performed using Kjeldahl Tablets ECO & Antifoam Tablets

Table 13 and 14 show the results of the samples that were digested using Kjeldahl Tablets ECO & Antifoam Tablets. The

relative standard deviations (RSD) of the blanks and the samples are fine. Moreover, the recovery rates of the reference samples are within the required range (98.0–102.0%) and the determined protein contents are very close to the expected values.

Compared to the standard method (KjelOptimizer App), up to an average of 39.5% NaOH per sample was saved for the first measurement series (Table 13). In total, the NaOH consumption was reduced by 28.1% (= 320 mL). For the second measurement series (Table 14, p. 13), up to 17.3% of

NaOH per sample was saved. In total, 9.4% (= 188 mL) of the NaOH was saved. However, up to 2.7% more NaOH in average was used for samples containing high volumes of conc. sulfuric acid after digestion (blanks & references).

Type		Titrated volume [mL]		N [%]		Ø Protein [%]		NaOH Vol. [mL]		Ø NaOH Vol. saved vs standard [%]	
Blank	-	0.302	0.3062	-	-	-	-	101	84.8 ± 13.5	-4.7	
		0.304						74			
		0.307						74			
		0.311						77			
		0.307						98			
Ref (Gly)	0.0959	6.679		18.65	-	-	99.96	104	92.0 ± 10.3	-13.6	
	0.0939	6.557		18.69			100.14	92			
	0.0988	6.885		18.69			100.16	85			
	0.0967	6.75		18.7			100.24	79			
	0.0971	6.761		18.66			100	100			
Sample No. 1	1.0764	12.711		3.23	20.64	20.65 ± 0.01		57	63.2 ± 12.8	22	
	1.1061	13.072		3.24	20.67			64			
	1.0313	12.198		3.24	20.65			85			
	1.1132	13.146		3.24	20.66			54			
	1.1053	13.059		3.24	20.66			56			
Sample No. 2	0.9585	17.422		5.01	31.98	31.88 ± 0.11		92	74.0 ± 13.7	8.6	
	0.9165	16.532		4.97	31.71			68			
	0.9522	17.25		4.99	31.87			61			
	0.9738	17.629		4.99	31.86			64			
	0.9319	16.964		5.02	32.01			85			
Sample No. 3	1.909	17.89		2.59	16.16	16.17 ± 0.02		63	74.6 ± 10.7	7.9	
	1.9866	18.621		2.59	16.17			74			
	1.9516	18.348		2.59	16.22			66			
	1.9727	18.476		2.59	16.16			81			
	1.9969	18.697		2.59	16.16			89			
Sample No. 4	1.2651	6.005		1.26	7.9	7.89 ± 0.01		55	56.2 ± 3.8	30.6	
	1.2885	6.099		1.26	7.89			53			
	1.3384	6.319		1.26	7.88			53			
	1.3597	6.423		1.26	7.89			58			
	1.3848	6.523		1.26	7.88			62			
Total NaOH Volume [mL]								2224			
Expected NaOH Volume without RDS (81 mL/sample) [mL]								2430			
NaOH Volume saved [mL]								206.0			
NaOH saved [%]								8.5			

Table 12: Results of the measurements performed using 20 mL of conc. H₂SO₄, Kjeldahl Tablets Missouri & Antifoam Tablets (No. 91-120; RSD: blank=1.00 %; ref=0.10 %; s 1=0.05 %; s 2=0.34 %; s 3=0.14 %; s 4=0.12 %).

Measurements performed using Kjeldahl Tablets ECO & Antifoam Tablets

Table 13 and 14 show the results of the samples that were digested using Kjeldahl Tablets ECO & Antifoam Tablets. The

relative standard deviations (RSD) of the blanks and the samples are fine. Moreover, the recovery rates of the reference samples are within the required range (98.0–102.0%) and the determined protein contents are very close to the expected values.

Type		Titrated volume [mL]		N [%]		Ø Protein [%]		NaOH Vol. [mL]		Ø NaOH Vol. saved vs standard [%]	
Blank	-	0.322	0.3198	-	-	-	-	27	31.4 ± 3.6	17.4	
		0.315						36			
		0.321						30			
		0.323						34			
		0.318						30			
Ref (Gly)	0.0981	6.814		18.58	-	-	99.88	28.8 ± 1.8	24.2		
	0.0935	6.547		18.69			100.49			29	
	0.0993	6.941		18.72			100.6			27	
	0.0923	6.467		18.69			100.49			27	
	0.0953	6.613		18.54			99.63			30	
Sample No. 1	0.4329	5.328		3.25	20.72	20.69 ± 0.02	-	30	26.6 ± 2.3	30	
	0.404	4.982		3.24	20.67			27			
	0.4838	5.908		3.24	20.69			24			
	0.4496	5.515		3.24	20.69			25			
	0.4816	5.878		3.24	20.67			27			
Sample No. 2	0.9956	17.203		4.76	30.37	30.64 ± 0.20	-	17	23.0 ± 6.0	39.5	
	0.9698	16.941		4.81	30.69			21			
	0.9527	16.697		4.83	30.78			22			
	0.944	16.383		4.78	30.47			22			
	0.9845	17.309		4.84	30.9			33			
Sample No. 3	1.6239	15.628		2.65	16.54	16.45 ± 0.06	-	23	27.2 ± 5.7	28.4	
	1.6496	15.803		2.63	16.47			27			
	1.6907	16.082		2.62	16.36			37			
	1.7391	16.585		2.63	16.41			24			
	1.6514	15.833		2.64	16.48			25			
Sample No. 4	0.6258	3.137		1.26	7.9	7.89 ± 0.02	-	26	27.0 ± 1.6	28.9	
	0.6226	3.108		1.26	7.86			27			
	0.697	3.461		1.27	7.91			28			
	0.6303	3.158		1.26	7.9			25			
	0.64	3.2		1.26	7.9			29			
Total NaOH Volume [mL]								820			
Expected NaOH Volume without RDS (38 mL/sample) [mL]								1140			
NaOH Volume saved [mL]								320			
NaOH saved [%]								28.1			

Table 13: Results of the measurements performed using 8 mL of conc. H₂SO₄, Kjeldahl Tablets ECO & Antifoam Tablets (No. 121–150; RSD: blank = 0.91%; ref = 0.39%; s 1 = 0.09%; s 2 = 0.64%; s 3 = 0.38%; s 4 = 0.22%).

Compared to the standard method (KjelOptimizer App), up to an average of 39.5% NaOH per sample was saved for the first measurement series (Table 13). In total, the NaOH consumption was reduced by 28.1% (= 320 mL). For the second measurement series (Table 14, p. 13), up to 17.3% of

NaOH per sample was saved. In total, 9.4% (= 188 mL) of the NaOH was saved. However, up to 2.7% more NaOH in average was used for samples containing high volumes of conc. sulfuric acid after digestion (blanks & references).

Type		Titrated volume [mL]		N [%]		Ø Protein [%]		NaOH Vol. [mL]		Ø NaOH Vol saved vs standard [%]
Blank	-	0.3130	0.3090	-	-	-	-	76	68.8 ± 5.0	-2.7
		0.3080						66		
		0.3100						72		
		0.3050						66		
		0.3090						64		
Ref (Gly)	0.095	6.5940	18.57	-	-	99.88	72	66.6 ± 3.7	0.6	
	0.0949	6.5780	18.54			100.49	62			
	0.0967	6.7100	18.58			100.6	68			
	0.0963	6.6960	18.62			100.49	66			
	0.0908	6.3300	18.61			99.63	65			
Sample No. 1	0.4732	5.7540	3.23	20.61	20.62 ± 0.01	-	57	66.6 ± 3.7	16.1	
	0.4635	5.6480	3.23	20.63			55			
	0.4484	5.4690	3.23	20.61			54			
	0.4614	5.6280	3.24	20.64			56			
	0.4789	5.8220	3.23	20.62			59			
Sample No. 2	0.9418	16.3330	4.78	30.47	30.77 ± 0.21	-	64	56.2 ± 1.9	10.7	
	0.9143	16.1590	4.87	31.05			56			
	0.9202	16.1990	4.85	30.92			51			
	0.913	15.9130	4.8	30.61			66			
	0.9675	16.9450	4.83	30.79			62			
Sample No. 3	1.9754	18.4720	2.58	16.13	16.14 ± 0.03	-	65	59.8 ± 6.2	17.3	
	1.9465	18.2240	2.58	16.15			50			
	1.9767	18.4430	2.58	16.09			53			
	1.961	18.3400	2.58	16.13			60			
	1.9851	18.6100	2.59	16.17			49			
Sample No. 4	1.0445	5.0110	1.26	7.90	7.90 ± 0.01	-	64	57.6 ± 10.3	14.0	
	1.059	5.0760	1.26	7.90			72			
	1.022	4.9110	1.26	7.90			50			
	1.1098	5.3090	1.26	7.90			47			
	1.0452	5.0080	1.26	7.90			55			
Total NaOH Volume [mL]								1822		
Expected NaOH Volume without RDS (38 mL/sample) [mL]								2010		
NaOH Volume saved [mL]								188		
NaOH saved [%]								9.4		

Table 14: Results of the measurements performed using 16 mL of conc. H₂SO₄, Kjeldahl Tablets ECO & Antifoam Tablets (No. 151–180; RSD: blank = 0.84%; ref = 0.15%; s 1 = 0.07%; s 2 = 0.68%; s 3 = 0.16%; s 4 = 0.07%).

Comparison Kjeldahl Distillation with & without Reaction Detection Sensor

Table 15 shows an overview of the distillation parameters/steps for a Kjeldahl distillation with and without RDS. Instead of three parameters (H₂O Volume, NaOH Volume & Reaction Time), only one parameter (H₂SO₄ Vol. for Digestion) needs to be defined in the method if the RDS is used. Based on the entered value, the system automatically performs a sample pre-conditioning and a water dosage. Furthermore, the end of the alkalization is detected and the dosing of NaOH is stopped automatically.

Distillation parameter/step	Without RDS	With RDS
Sample pre-conditioning	Not performed	Performed
H ₂ SO ₄ Vol. for digestion	Not required	Required (8–30 mL)
H ₂ O Volume	Required (xx mL)	Not required (automatically closed)
NaOH Volume	Required (xx mL)	Not required (dosage stops automatically after detection)
Reaction Time	Required (xx s)	Not required

Table 15: Comparison Kjeldahl method programming with and without RDS.

CONCLUSION

The determination of nitrogen/protein in milk powder, Emmentaler cheese, boiled sausage and milk chocolate using the KjelDigester K-449 and KjelLine systems with the Reaction Detection Sensor provides reliable and reproducible results. All results correspond well to the labelled values with low relative standard deviations. Furthermore, all recovery rates for Glycine are within the required range (98.0–102.0%).

Due to the RDS technology, sodium hydroxide solution was saved in all measurement series. The RDS shows the best performance for samples digested with Kjeldahl Tablets Titanium. In those cases, an average NaOH saving of up to

43.2% per sample was reached with a very reproducible detection of the alkalization endpoint. For samples containing Kjeldahl Tablets Missouri or ECO, the performance is less good. Compared to a Kjeldahl distillation without RDS, even more sodium hydroxide is dosed in some cases, especially if a high volume of sulfuric acid remains in the sample tube after digestion (e.g. blank, reference or sample with little organic matter). Nevertheless, the RDS can also be used with these Kjeldahl Tablets to significantly reduce sodium hydroxide consumption. At low sulfuric acid volumes for digestion (e.g. 8 or 10 mL), sodium hydroxide was saved in all measurements. In addition, the sodium hydroxide consumption is significantly reduced for samples containing a high amount of organic matter and therefore, consumed a lot of sulfuric acid during digestion.

REFERENCES

- [1] Technical Note No. 772: Reaction Detection Sensor
- [2] Kjeldahl Optimizer App
- [3] Kjeldahl Knowledge Base
- [4] Operation Manual of KjelDigester K-449
- [5] Operation Manual of Scrubber K-415
- [6] Operation Manual of Kjel Line
- [7] Operation Manual of Dist Line
- [8] Application Note – N°. 777/2021

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Kjel Line

Meeting the highest demands in nitrogen determination

